SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: MLA03
COURSE CODE: Reinforcement learning

COURSE OUTCOME COVERED

CO5: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 35%

Dr G .A. SENTHIL

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Scenario-Based Assessment **DURATION: 1.5 Hrs**

1. A mobile app company wants to improve user engagement by showing personalized notifications at the right time. The company plans to use Reinforcement Learning to learn user behavior patterns. Report how the RL framework can be applied by defining the possible states, actions, and reward functions for this scenario.
(5 Marks)

2. A warehouse robot uses Reinforcement Learning to find the shortest path for delivering packages while avoiding obstacles. Sometimes the robot chooses longer paths due to insufficient exploration during training. Solve how the exploration vs exploitation trade-off affects the robot's navigation performance and suggest some improvements. **(5 Marks)**

3. A hospital is considering using Reinforcement Learning to optimize patient appointment scheduling in order to reduce waiting time and improve resource utilization. Demonstrate the suitability of RL for this scenario and discuss possible challenges such as delayed rewards and ethical considerations. **(5 Marks)**

4. A smart home system aims to reduce electricity consumption by automatically controlling lights, fans, and air conditioners based on user presence and weather

conditions. Sketch a Reinforcement Learning model for this scenario by defining state space, action space, reward structure, and suitable learning algorithm.

(5 Marks)

5. An online gaming platform uses Reinforcement Learning to match players of similar skill levels to maintain engagement. Report how reward design can influence matchmaking quality with reward definition that might lead to biased or inefficient matching results. **(5 Marks)**

6. A financial trading system applies Reinforcement Learning to decide buy/sell actions based on market signals. Analyze how states, actions, and rewards can be defined in this context and evaluate the risks of delayed rewards in volatile markets.

(5 Marks)

7. A drone delivery company uses RL to optimize delivery routes while avoiding restricted airspace. Construct a suitable RL framework by defining states, actions, and rewards, and justify how penalties can ensure safe navigation. **(5 Marks)**

8. A university applies RL to personalize online learning paths for students. Examine how reward design can influence student engagement and defend the ethical considerations of adapting learning content dynamically. **(5 Marks)**

9. A traffic management authority deploys RL to control traffic lights at busy intersections. Illustrate how constraints such as emergency vehicle priority and pedestrian safety can be incorporated into the RL model and interpret the impact on congestion reduction. **(5 Marks)**

10. A manufacturing plant uses RL to schedule machines under resource limitations. Differentiate between traditional optimization methods and constrained RL approaches, and assess which method provides better adaptability in dynamic production environments. **(5 Marks)**

Code of Conduct:

I ______M.Vineela(192472071)______________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

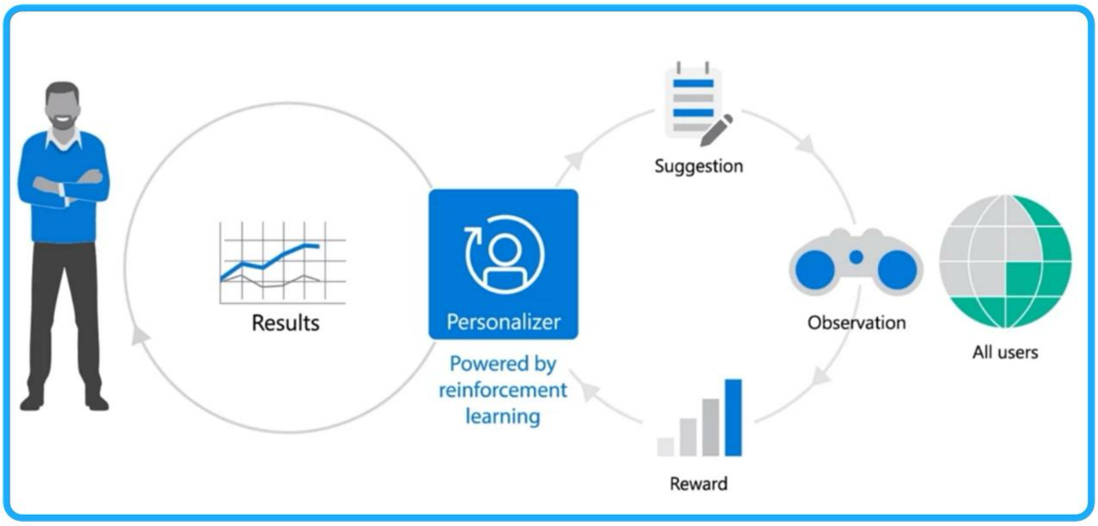
Signature of the student:

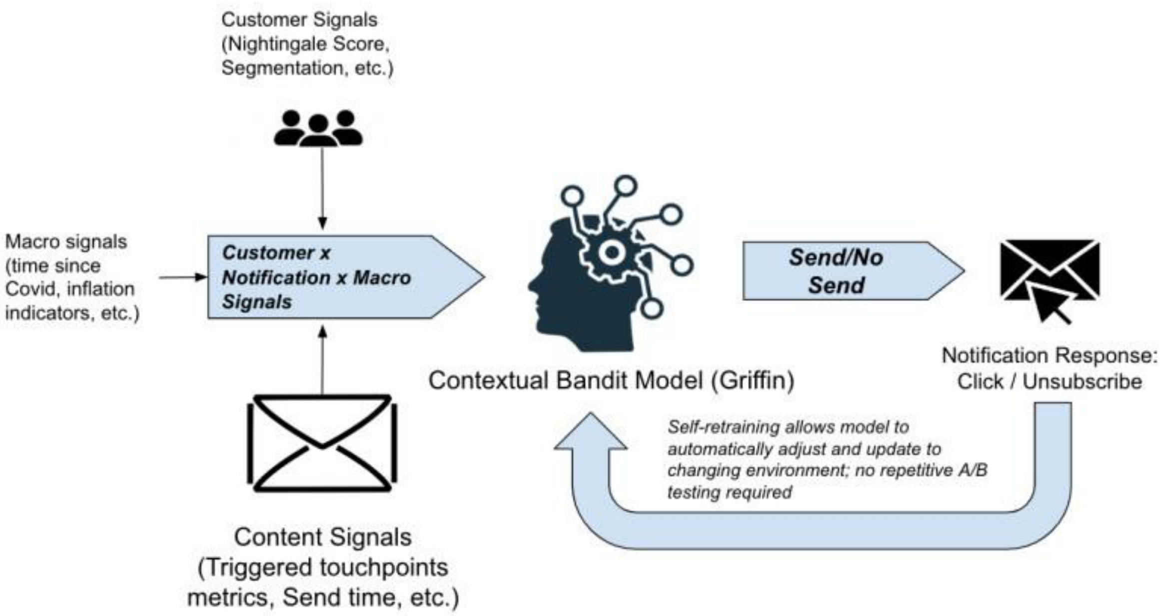
RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Reinforcement Learning – Application-Based Questions

1. Personalized Notifications Using Reinforcement Learning



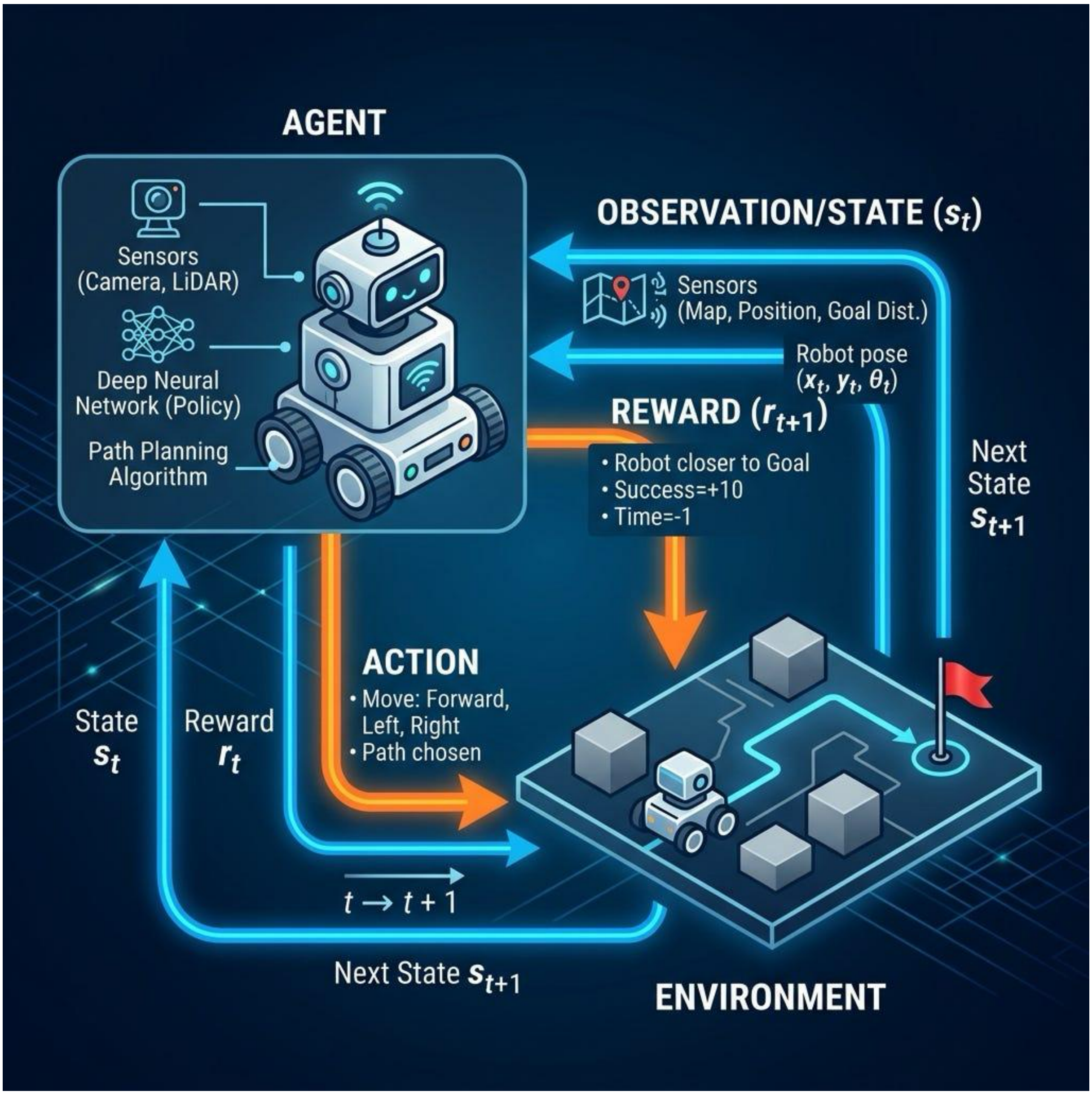


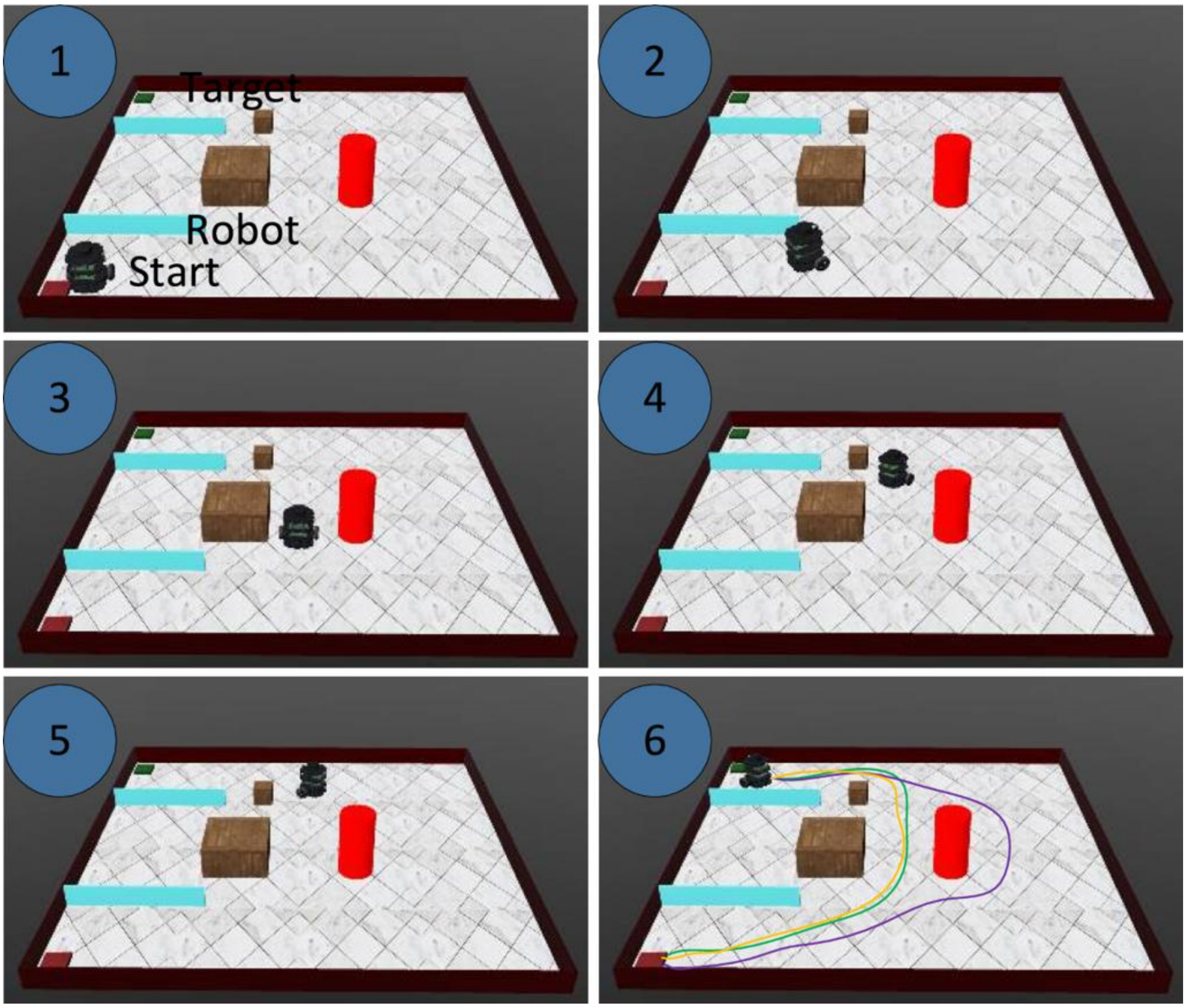
A mobile application can use Reinforcement Learning (RL) to learn the best time and type of notification for each user.

- **State:** The state can include user's activity level, time of day, previous notification responses, app usage frequency, location context, and recent interactions.
- **Actions:** The agent can choose to send a notification, delay it, or select different notification types such as promotional, reminder, or personalized content.
- **Reward:** A positive reward can be given when the user opens, clicks, or interacts with the notification. A negative reward can be assigned when the user ignores, dismisses, or disables notifications.
- **Learning:** The RL agent observes user responses and continuously updates its policy to determine the best notification timing.
- **Objective:** The system maximizes long-term user engagement while avoiding excessive notifications.

Thus, RL allows the application to provide adaptive and personalized notifications based on individual user behavior.

2. Exploration vs Exploitation in Warehouse Robot Navigation




1
Target
Robot
Start
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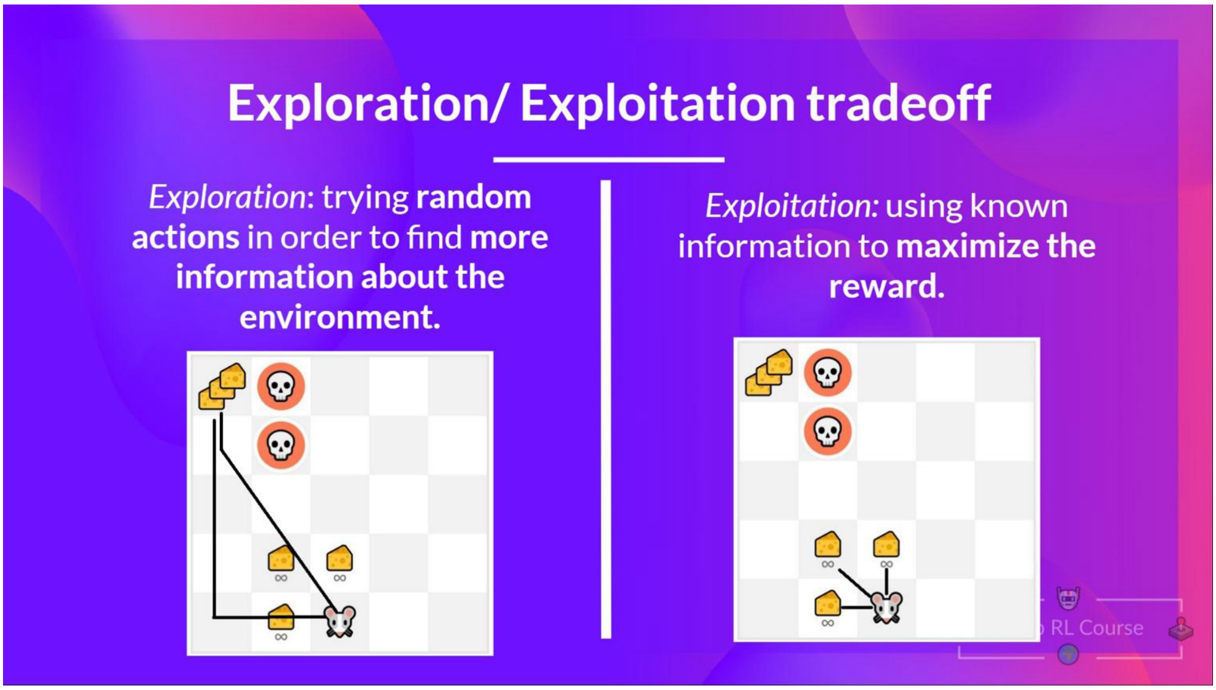

Exploration/ Exploitation tradeoff
Exploration: trying **random actions** in order to find **more information about the environment.**
Exploitation: using known information to **maximize the reward.**
RL Course

The exploration-exploitation trade-off determines how the warehouse robot learns efficient paths.

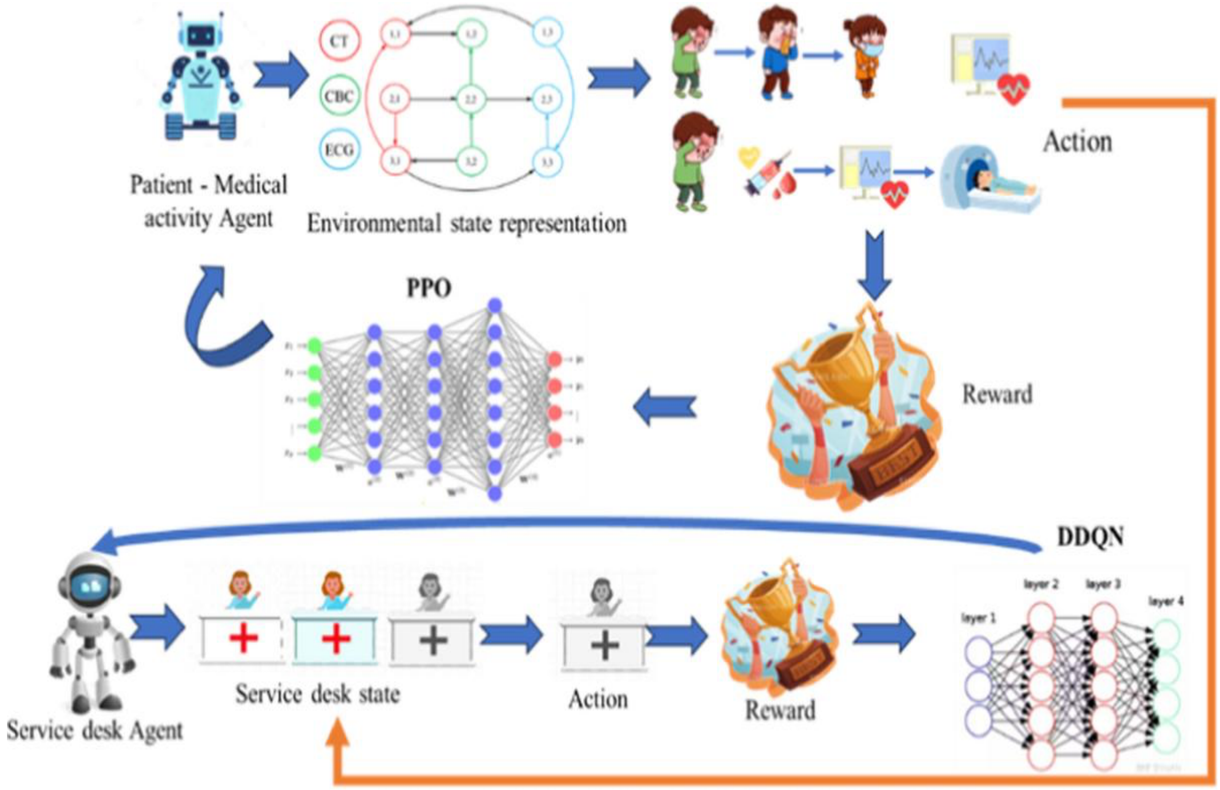
- **Exploration:** The robot tries different routes to discover potentially shorter and safer paths.
- **Exploitation:** The robot uses the best path it has already learned.
- If exploration is **too low**, the robot repeatedly selects a known longer path and may never discover a shorter route.
- If exploration is **too high**, the robot wastes time trying inefficient paths and may increase delivery time.
- An **ε-greedy strategy** can balance both: with probability ε the robot explores a new action, while with probability it selects the best-known action.

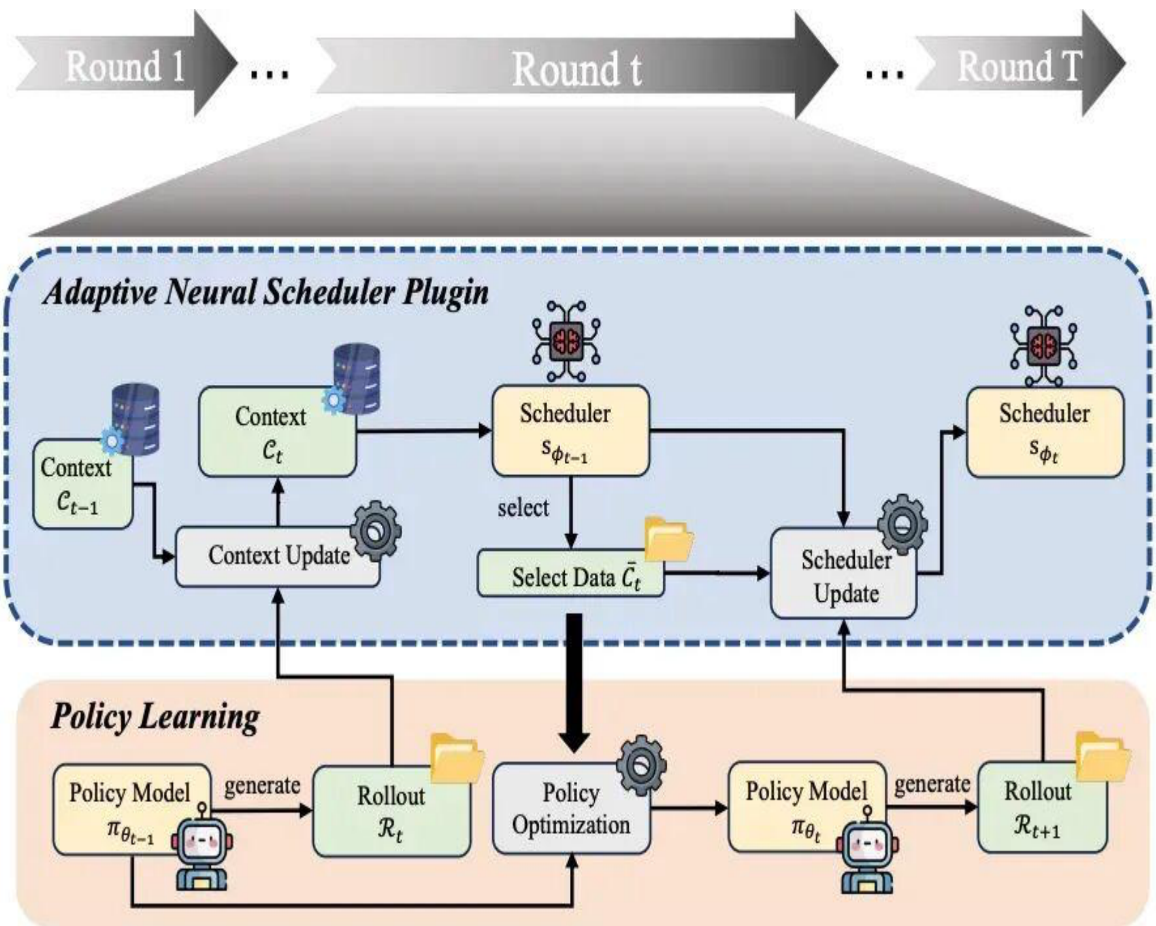
Improvements

1. Start with a high exploration rate and gradually decrease it.
2. Use **ε-decay** during training.
3. Apply **Q-learning or Deep Q-Network (DQN)** for complex environments.
4. Give higher rewards for shorter paths and penalties for collisions.
5. Use experience replay to learn from previous navigation experiences.

Therefore, balanced exploration helps the robot discover efficient paths while exploitation allows it to use learned routes effectively.

3. RL for Hospital Appointment Scheduling



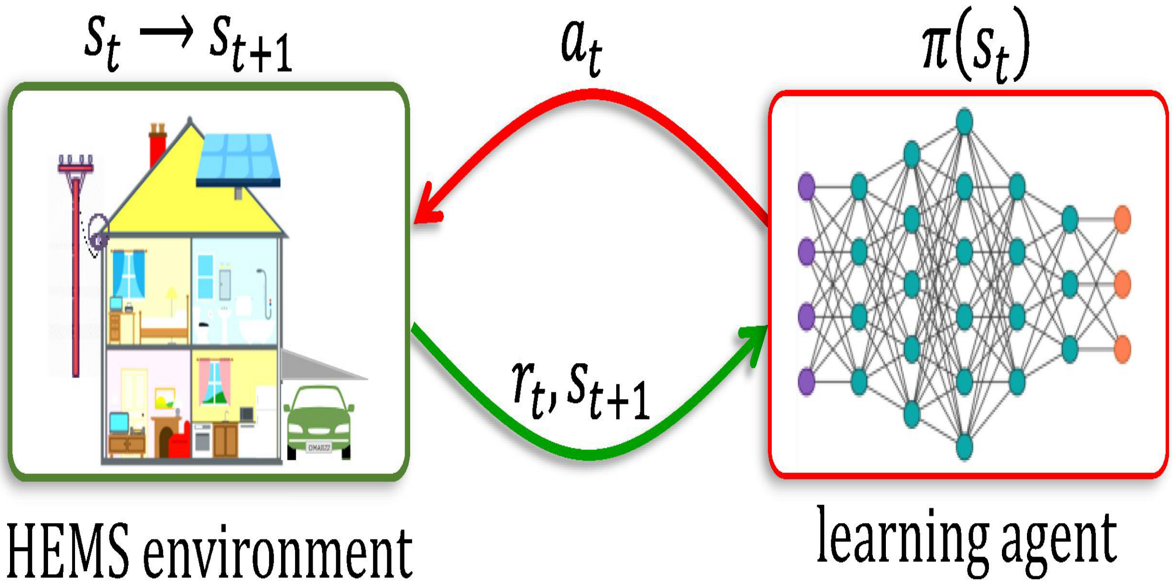


RL is suitable because appointment scheduling is a sequential decision-making problem where decisions affect future waiting times and resource utilization.

- **State:** Number of waiting patients, doctor availability, room availability, appointment types, patient priority, and current time.
- **Actions:** Assign a patient to a doctor, select an appointment time, reschedule an appointment, or allocate resources.
- **Reward:** Positive reward can be provided for reducing waiting time and improving resource utilization. Penalties can be given for long waiting times, cancellations, and resource underutilization.
- **Delayed rewards:** The effect of an appointment decision may only become visible later when patients arrive and queues develop.
- **Ethical challenges:** The system must avoid discrimination, protect patient privacy, maintain fairness, and ensure that urgent patients receive appropriate priority.
- **Safety:** Human administrators should be able to override RL decisions when necessary.

Thus, RL can improve scheduling efficiency, but healthcare deployment requires strong safety, fairness, privacy, and human supervision.

4. Smart Home Energy Management Using RL



A smart home can use RL to automatically control electrical appliances while maintaining user comfort.

State Space

The state may contain:

- User presence/absence
- Indoor temperature
- Outdoor temperature
- Weather conditions
- Current electricity consumption
- Time of day
- Status of lights, fans, and AC

Action Space

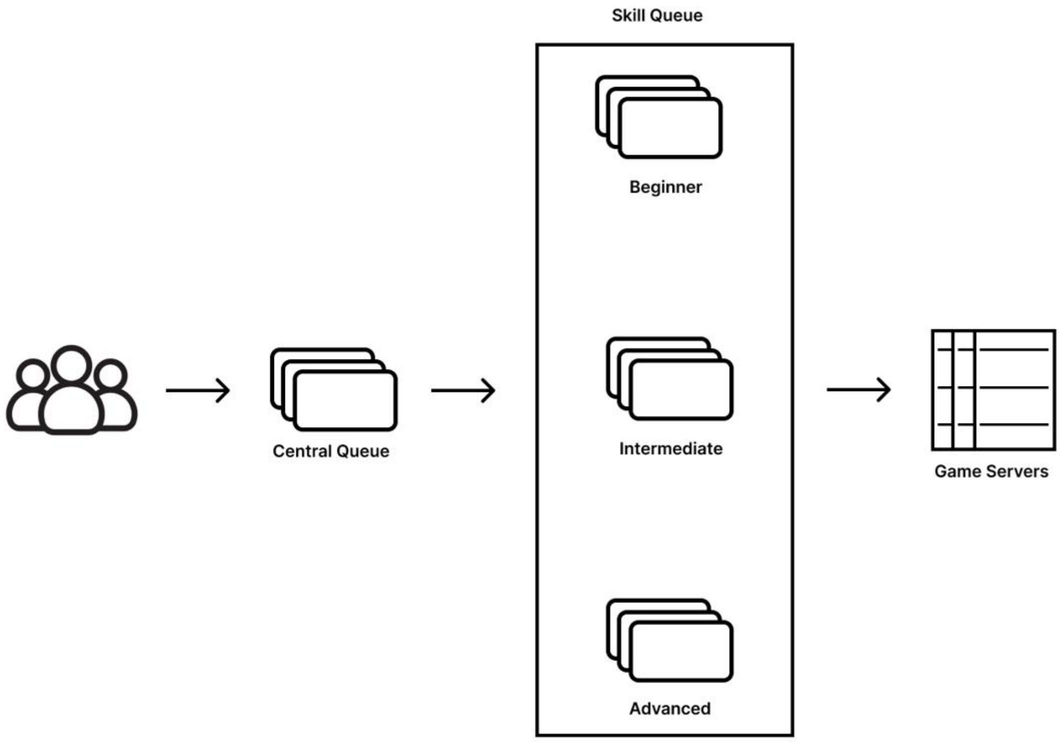
The agent can:

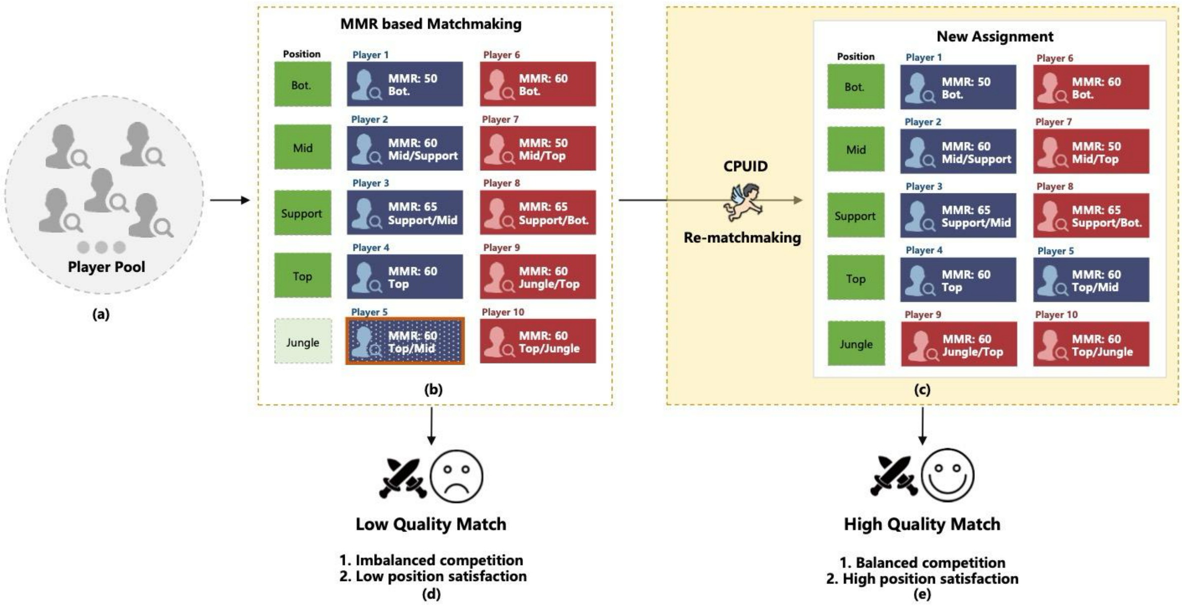
- Turn lights ON/OFF
- Adjust fan speed
- Turn AC ON/OFF
- Change AC temperature
- Reduce appliance usage

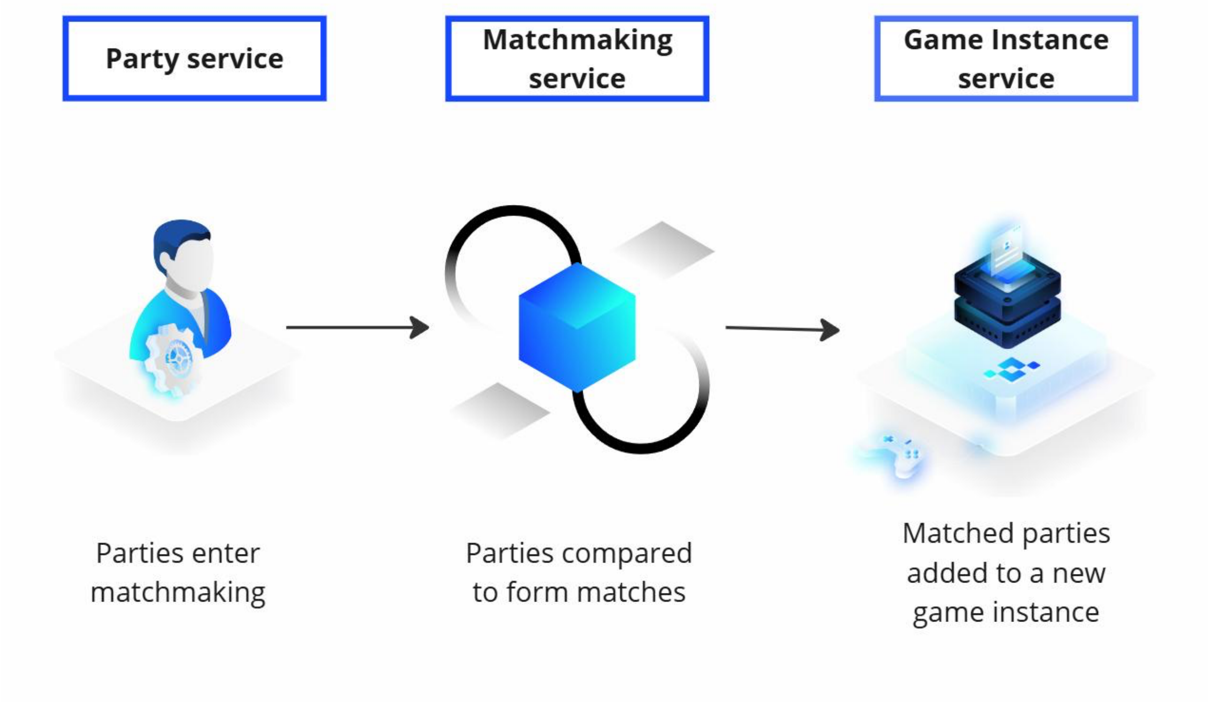
Reward Structure: A suitable reward function is:

A positive reward is given for maintaining comfort with low energy consumption, while penalties are given for excessive electricity usage or uncomfortable temperatures.

5. RL-Based Player Matchmaking





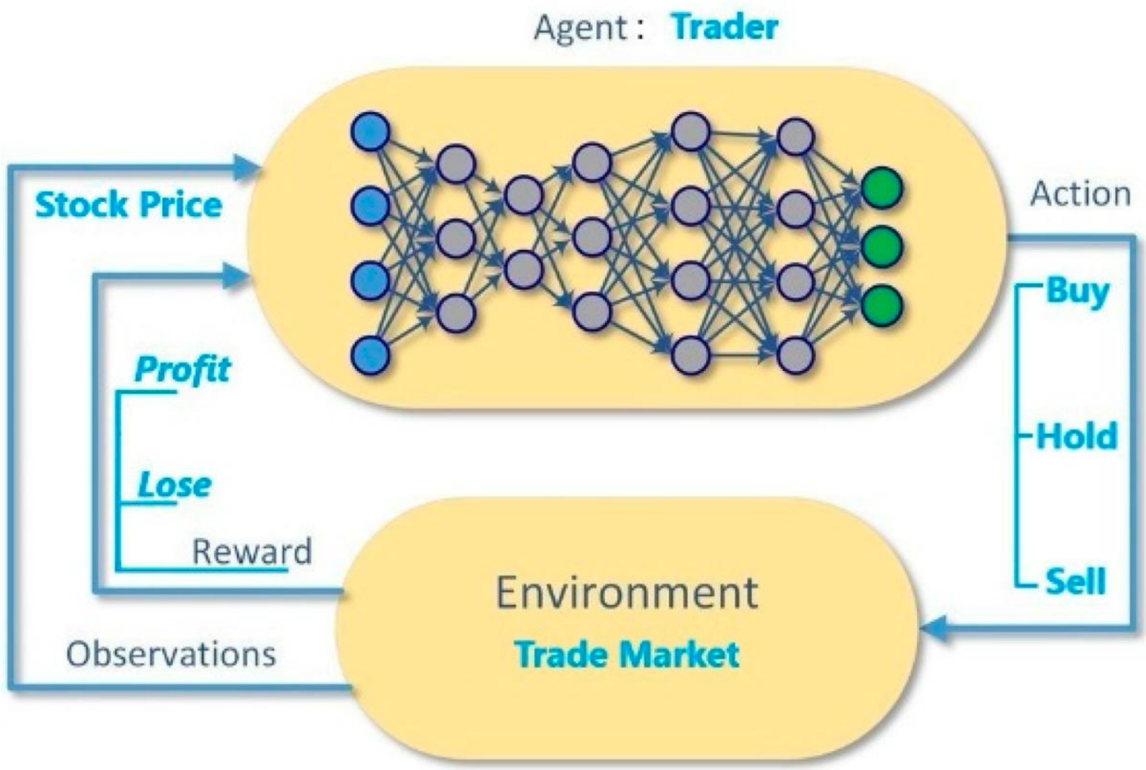


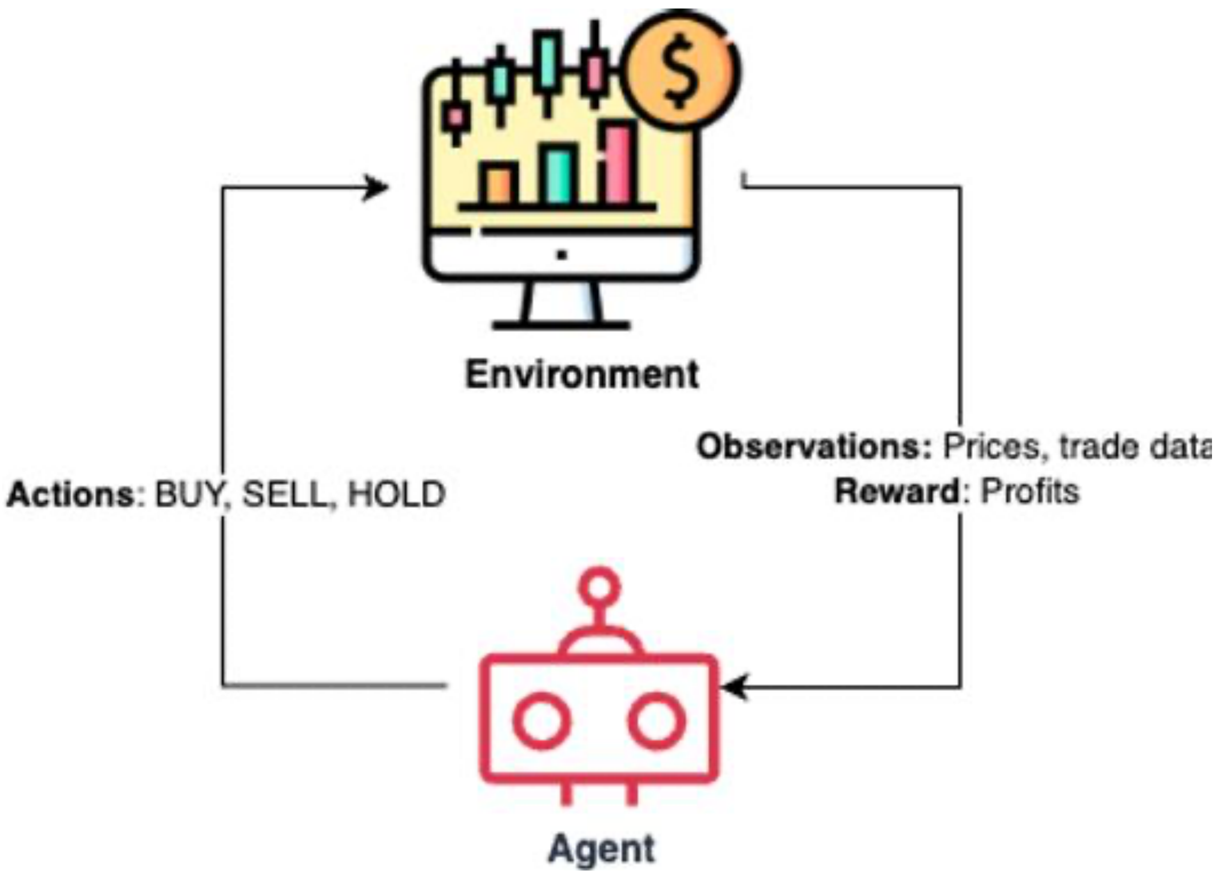
Reward design strongly influences the quality of player matchmaking.

- **State:** Player skill rating, experience, recent performance, win/loss history, latency, and waiting time.
- **Action:** Match a player with another player or group having a particular skill level.
- **Good reward:** Reward the agent when players have balanced matches, remain engaged, and report good gaming experiences.
- **Poor reward:** If the system rewards only quick matchmaking, it may match players with highly different skill levels simply to reduce waiting time.
- If reward is based only on **winning**, the system may repeatedly match strong players together, creating poor experiences for weaker players.
- A balanced reward can consider **fairness** + **engagement** + **waiting time** + **skill similarity**.

Therefore, carefully designed rewards prevent biased and inefficient matchmaking and encourage competitive but enjoyable games.

6. Reinforcement Learning for Financial Trading





RL can model financial trading as a sequential decision-making problem.

State

The state can include:

- Stock price
- Trading volume
- Technical indicators
- Moving averages
- Market trends
- Current portfolio balance
- Existing holdings

Actions

The agent can:

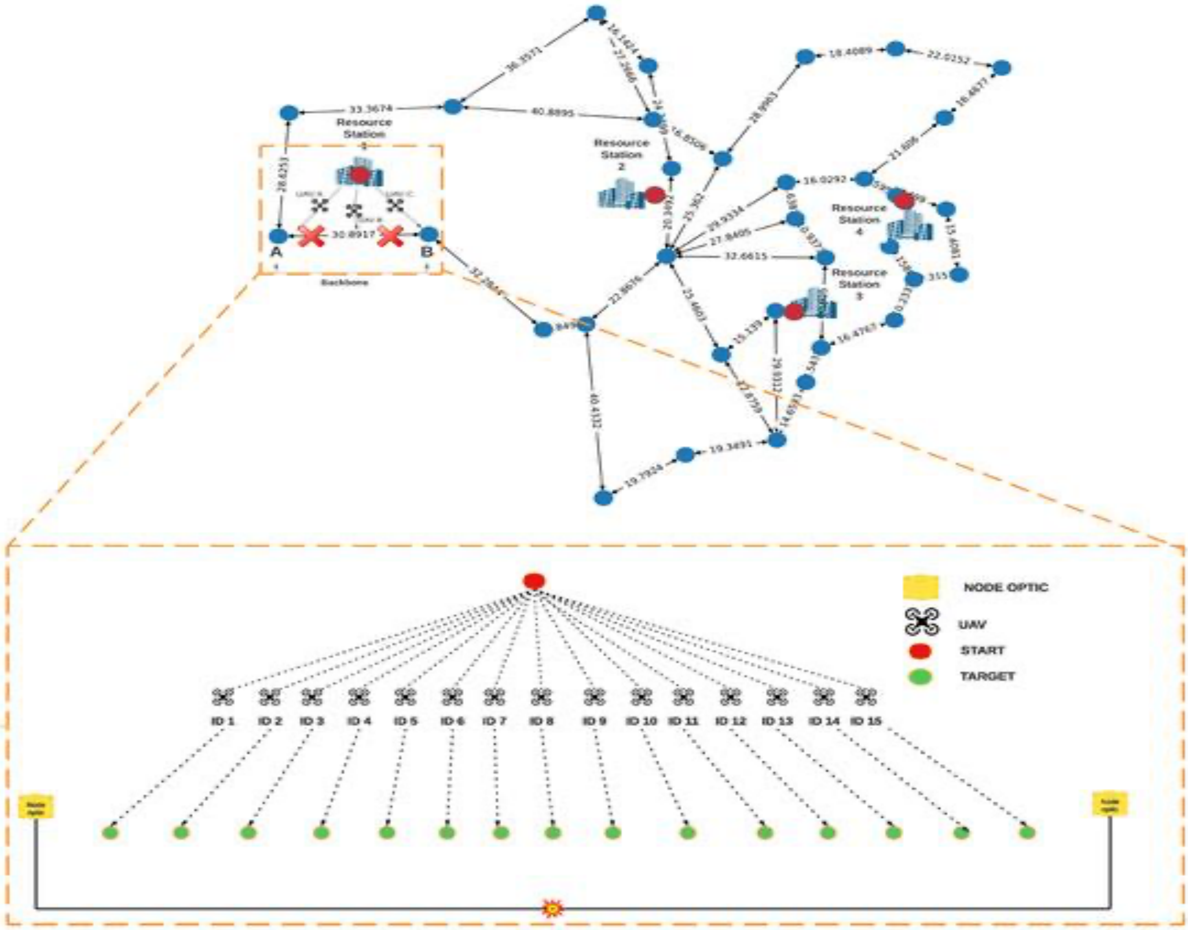
- **Buy**
- **Sell**
- **Hold**

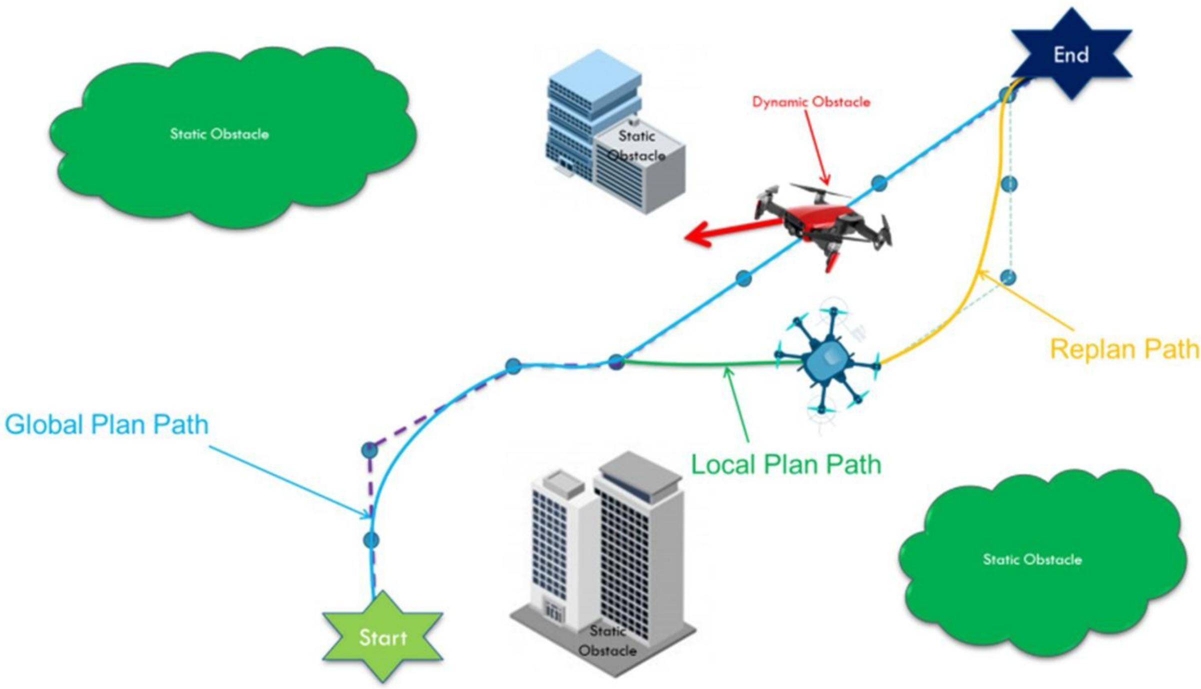
Reward Other risks include:

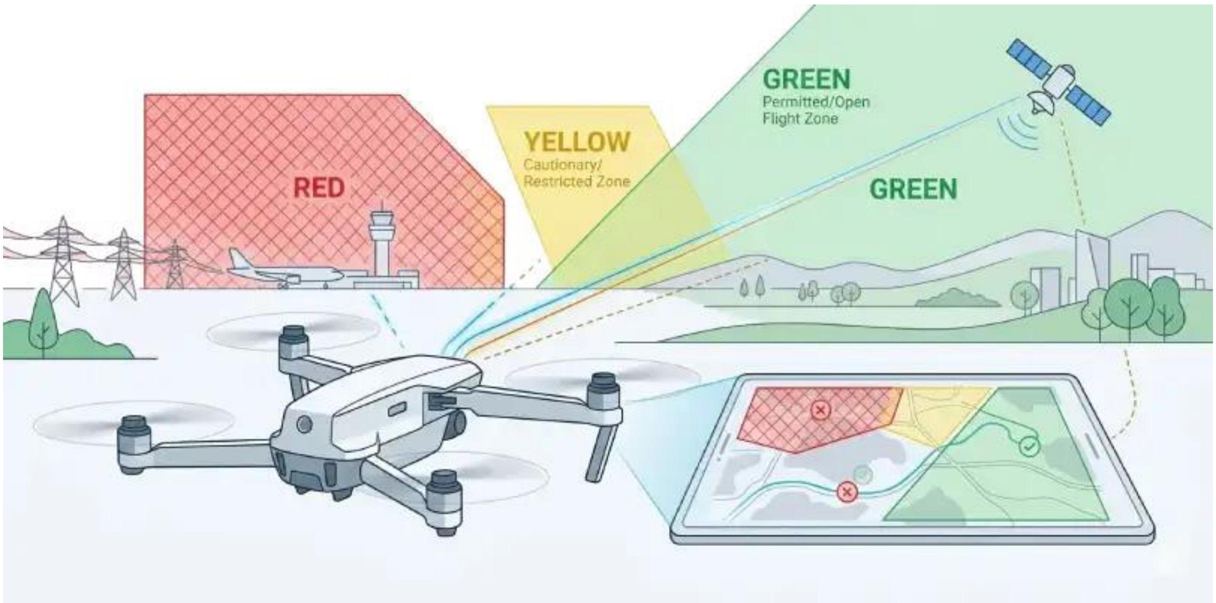
- Market volatility
- Financial loss
- Overfitting historical data
- Unexpected market events
- Transaction costs

Therefore, RL can support trading decisions, but strong risk management and testing are essential.

7. RL for Drone Delivery Route Optimization







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An RL-based drone delivery system can learn safe and efficient routes.

- **State:** Drone GPS position, altitude, battery level, destination, obstacles, weather conditions, and restricted-airspace information.
- **Actions:** Move forward, backward, left, right, change altitude, hover, or select another route.
- **Reward:** Positive reward for reaching the destination quickly and efficiently.
- **Penalties:** Large negative rewards can be assigned for entering restricted airspace, collision, excessive battery usage, or unsafe altitude.
- **Safety constraints:** Restricted zones can be represented as forbidden states or actions.
- **Algorithm:** DQN or PPO can be used depending on the complexity and continuous nature of drone movement.

A strong penalty for unsafe actions makes the agent learn that violating airspace restrictions is much worse than taking a longer safe route.

8. RL for Personalized Online Learning

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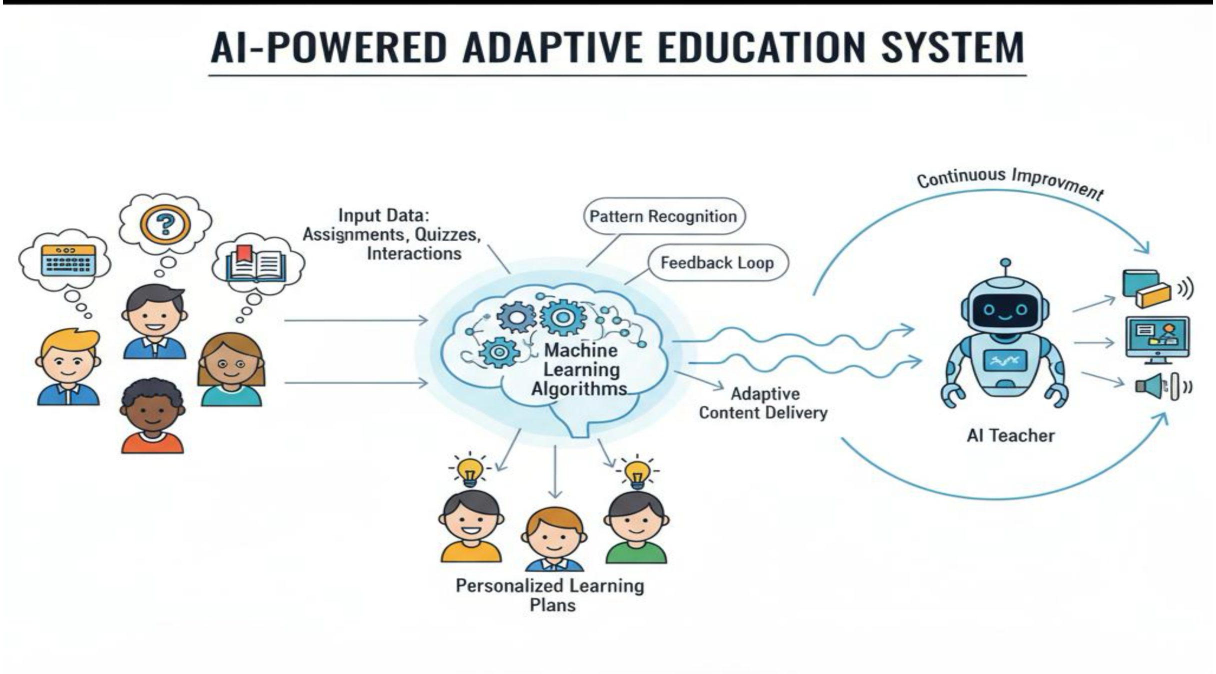

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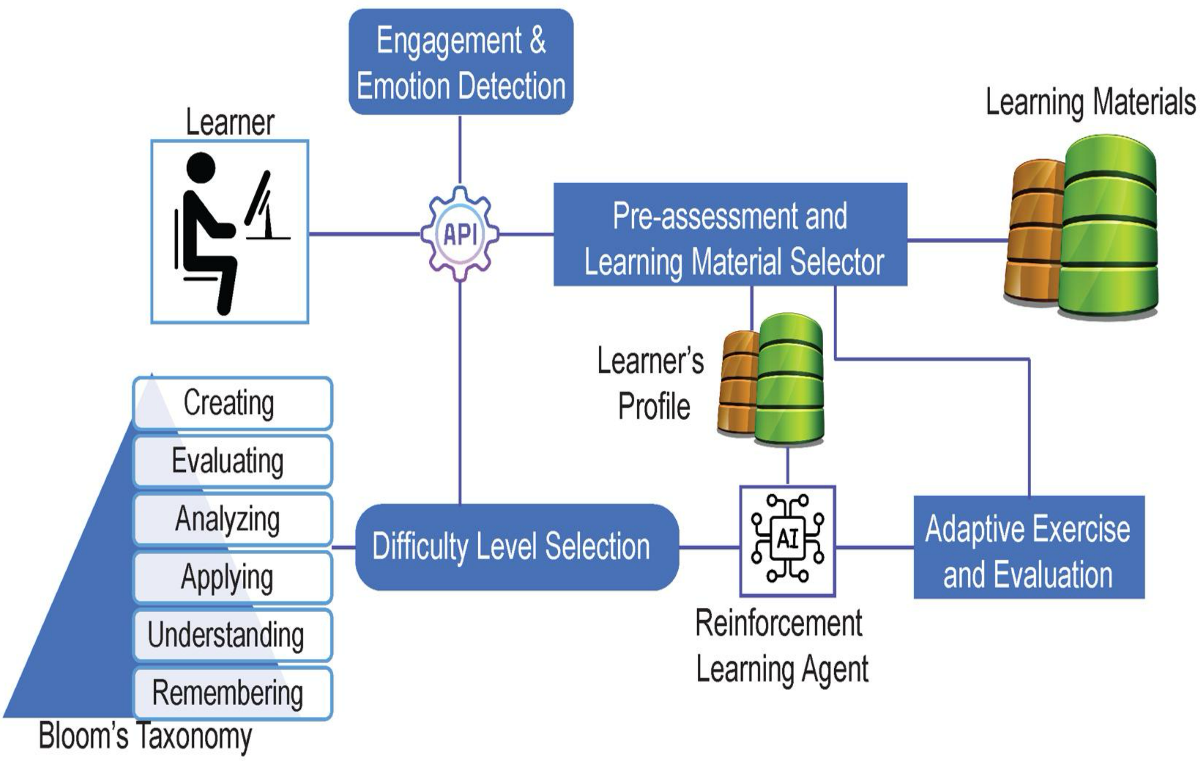


RL can personalize learning by selecting content according to a student's progress.

- **State:** Student knowledge level, quiz scores, learning speed, completed topics, mistakes, and engagement.
- **Actions:** Select a new lesson, revision activity, quiz, video, or difficulty level.
- **Reward:** Rewards can be based on learning improvement, quiz performance, course completion, and meaningful engagement.
- A poorly designed reward that focuses only on **time spent online** may encourage the system to show easy or entertaining content rather than useful educational material.
- The reward should therefore prioritize learning outcomes, engagement, progress, and appropriate difficulty.

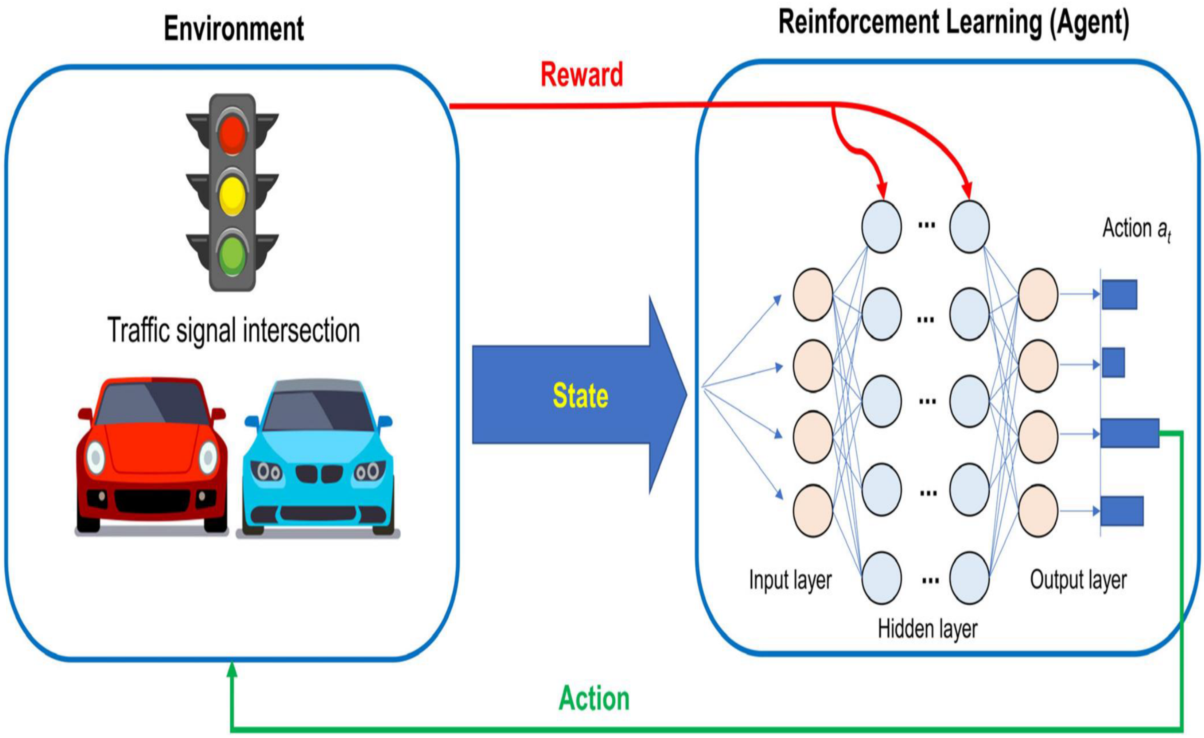
Ethical Considerations

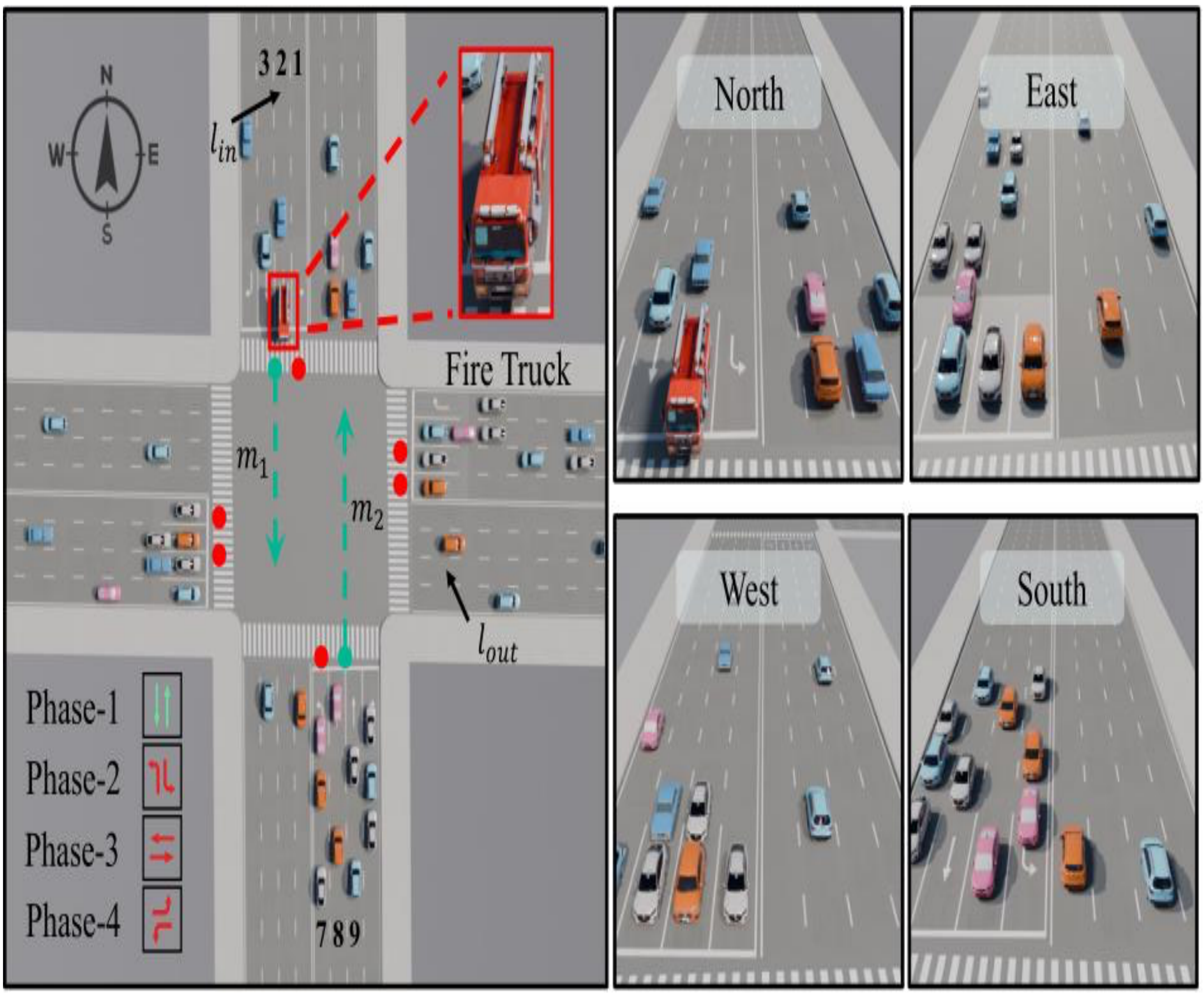
The system should:

1. Protect student data and privacy.
2. Avoid unfairly labeling students.
3. Provide equal learning opportunities.
4. Avoid excessive personalization that limits exposure to useful content.
5. Allow teachers and students to understand or override recommendations.

Thus, RL can provide adaptive education while requiring fairness, transparency, and privacy protection.

9. RL for Traffic Light Control



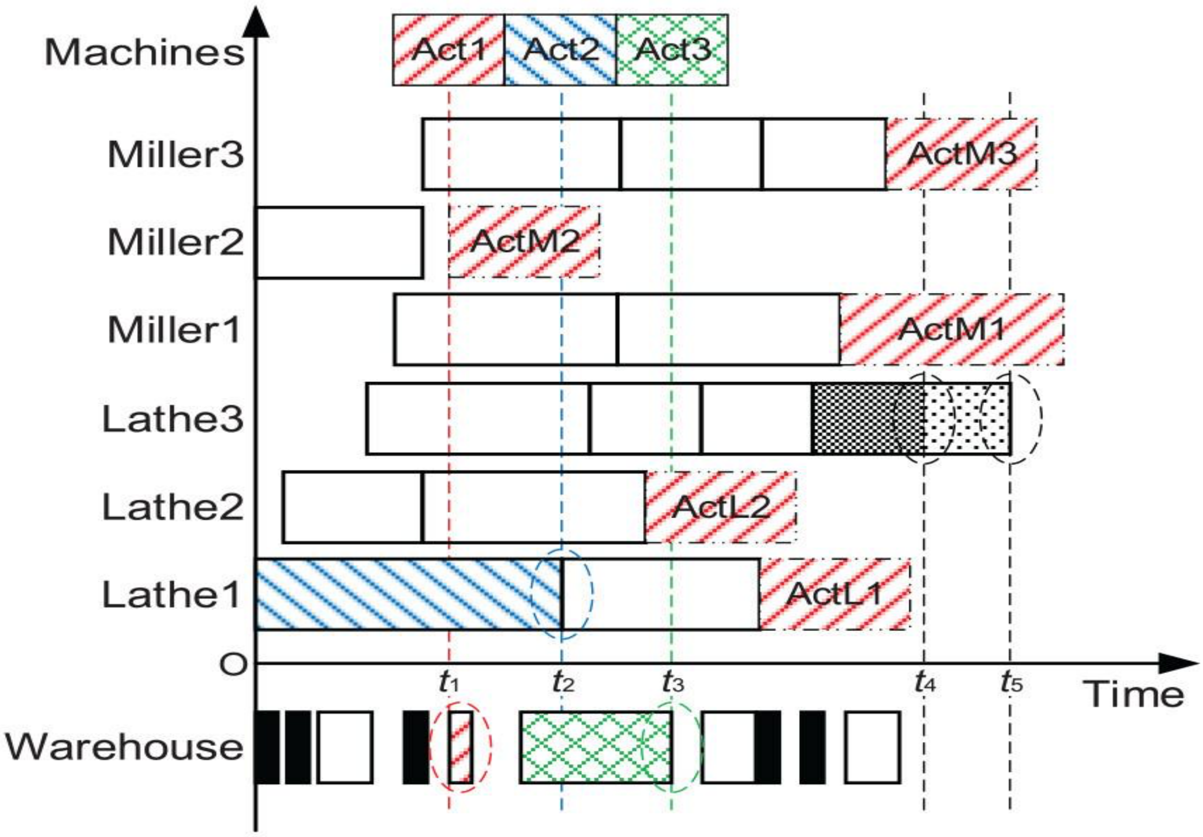


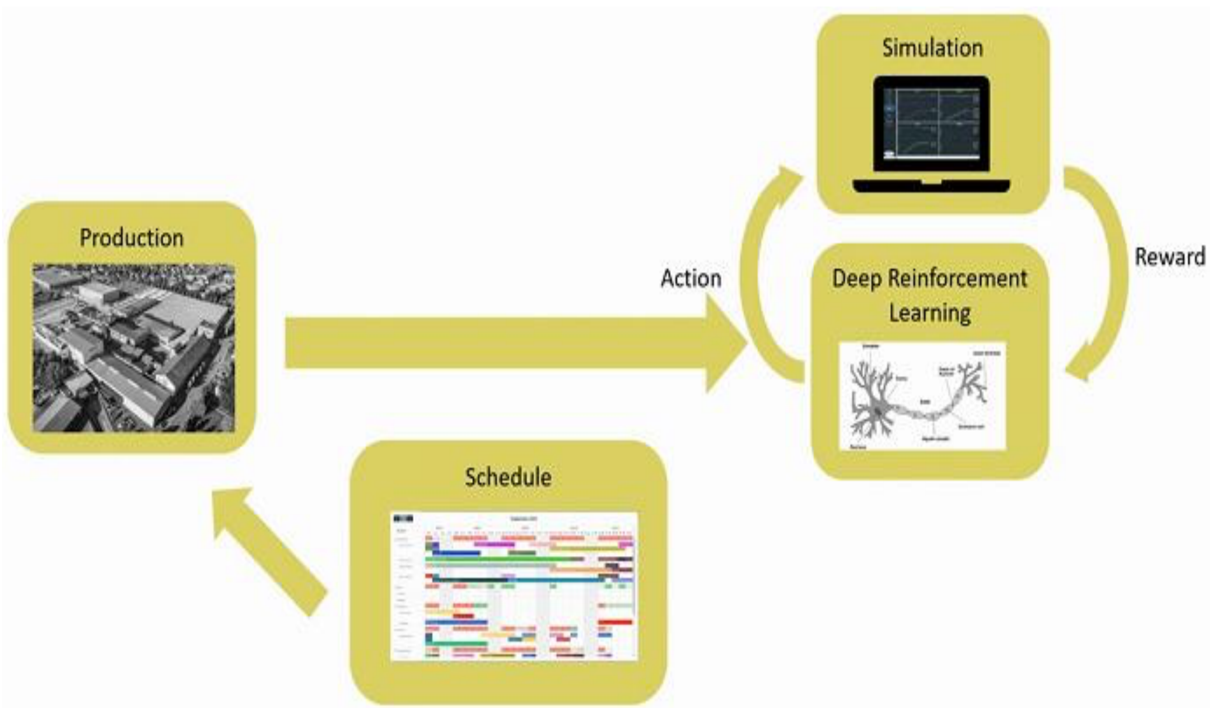
RL can dynamically control traffic signals according to current traffic conditions.

- **State:** Number of vehicles, queue length, waiting time, traffic flow, pedestrian crossing requests, and emergency vehicle presence.
- **Actions:** Change traffic-light phases, extend green time, reduce green time, or switch to another phase.
- **Reward:** Negative reward can be assigned for long queues and waiting times. Positive reward can be given for smooth traffic flow.
- **Emergency priority:** If an ambulance or fire truck is detected, the system can give priority by temporarily changing the signal.
- **Pedestrian safety:** Minimum pedestrian crossing time and safe signal phases can be imposed as hard constraints.
- **Congestion reduction:** The RL agent learns which signal timings reduce vehicle queues and average waiting time.

Therefore, RL can outperform fixed-time signals in changing traffic conditions while safety constraints prevent unsafe signal decisions.

10. RL vs Traditional Optimization in Machine Scheduling





Traditional optimization and constrained RL differ in how they handle production changes.

Aspect	**Traditional Optimization**	**Constrained RL**
Decision method	Solves a predefined optimization problem	Learns a policy through interaction
Dynamic changes	May require recalculation	Can adapt continuously
Resource constraints	Explicitly modeled	Included as constraints or penalties
Learning	Usually does not learn from experience	Improves through experience
Response to uncertainty	Limited depending on method	Better adaptability
Examples	Linear programming, MILP, genetic algorithms	PPO, DQN, constrained RL

Traditional Optimization: Methods such as Mixed Integer Linear Programming (MILP) can produce excellent schedules when the production environment and constraints are known. However, frequent machine failures, changing orders, and resource availability may require repeated optimization.

Constrained RL: Constrained RL learns scheduling policies while considering limitations such as machine capacity, labor availability, processing time, and deadlines.

Assessment: For a stable production environment, traditional optimization may provide highly accurate solutions. For a dynamic production environment, constrained RL generally provides better adaptability because it can learn from changing conditions and respond to new situations.